

CyberDefense Pro 2.0

2.5.2 Configuration Management

This lesson covers the following topic:

- Centralized operating system, application, and device management

Centralized Operating System, Application, and Device Management

Centralized operating system, application, and device management is a process that allows for the management of multiple systems, applications, and devices from a single location. This allows administrators to easily manage, update, and configure various systems, applications, and devices on the network from a central server instead of performing these tasks manually. Centralized operating system, application, and device management also allow for automated patching and configuration management processes, ensuring that all systems, applications, and devices are updated with the latest patches and configurations.

A Centralized Configuration Management System (CCMS) should be used to implement this. A centralized configuration management system allows an administrator to define device configuration settings on a management server and then push the settings to endpoints in an automated way.

Centralizing configurations enables consistency as the configuration is defined once and applied to many systems. Additionally, the configuration is enforced, meaning the central management server will overwrite changes made to an individual endpoint's configuration. As a result, the central management server can provide near real-time visibility into configuration changes. Near-real-time visibility into device configurations enables continuous compliance monitoring. When a secure configuration is centrally configured and controlled in this way, any changes to it (on an endpoint) will generate an immediate alert.

Examples of configuration management tools include the following:

- Chef - <https://www.chef.io/>
- Puppet - <https://puppet.com/>
- Ansible - <https://www.ansible.com/>

- Terraform - <https://www.terraform.io/>

The following video discusses the importance of configuring and implementing endpoint security controls:

Copyright © CompTIA, Inc. All rights reserved.